University of Nottingham
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Applications of AI in EEE

EEEE3134

Modern Programming

Git/GitHub and AI Agent

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Git/GitHub

- **What is Git/GitHub?**
  - A distributed version control system that **tracks changes to files and source code over time**.
- **Why Use Git/GitHub?**
  - Track changes to your code
  - Collaborate with multiple developers
  - Restore previous versions
  - Manage different features using branches
  - Backup projects through remote repositories
- **Workflow**

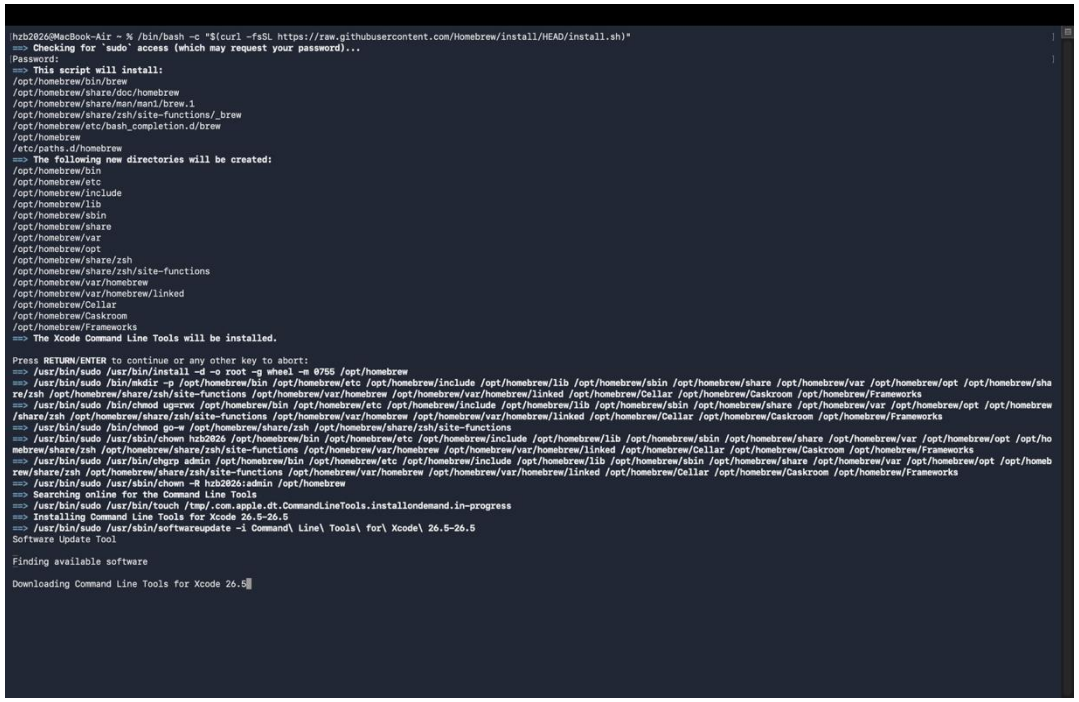


For a New Computer w/ macOS

- **Install Git**
  - **Download** and **install** it from official website: *https://git-scm.com/install*



*First install **Homebrew**:*
---- most popular package manager for macOS like App Store



For a New Computer w/ macOS



- **Use Git**
  - **Verification**: *git --version*
  - **Config Git**
    - *git config --global init.defaultBranch main*
    - *git config --global user.name "Zhengbing He"*
    - *git config --global user.email "he.zb@hotmail.com"*
  - **Initialize** a **Repository**
    - *git init*
  - **Check Repository Status**: *git status*

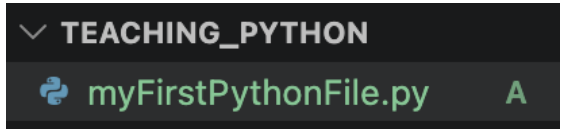
```
hzb2026@MacBook-Air teaching_python % git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        myFirstPythonFile.py
```

For a New Computer w/ macOS

- **Use Git**
  - **Move file** to **staging area**: *git add *.py*



A = Added
staged for commit

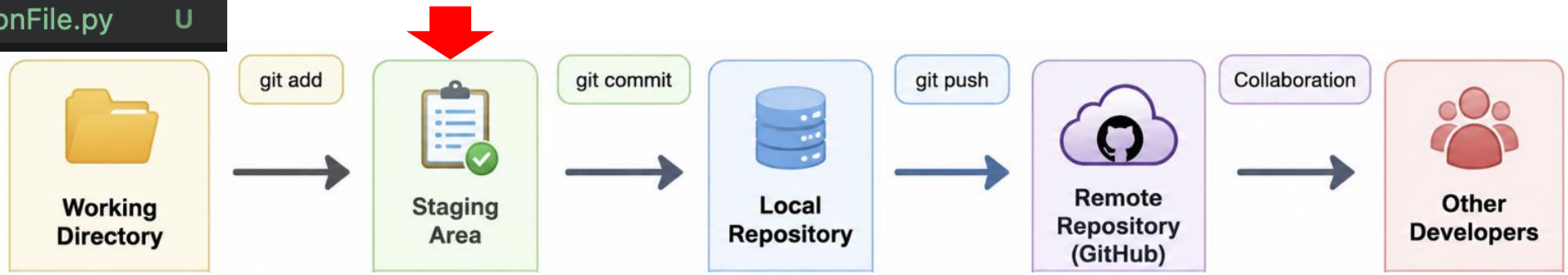
  - **Save staging files** to **local repository**: *git commit -m "Initial commit"*

Short and descriptive commit message
- *"Add login page"*
- *"Fix calculation error"*
- *"Update README"*

```
[main (root-commit) c207159] Initial Commit
 1 file changed, 1 insertion(+)
 create mode 100644 myFirstPythonFile.py
```

U = Untracked *not tracked by Git*

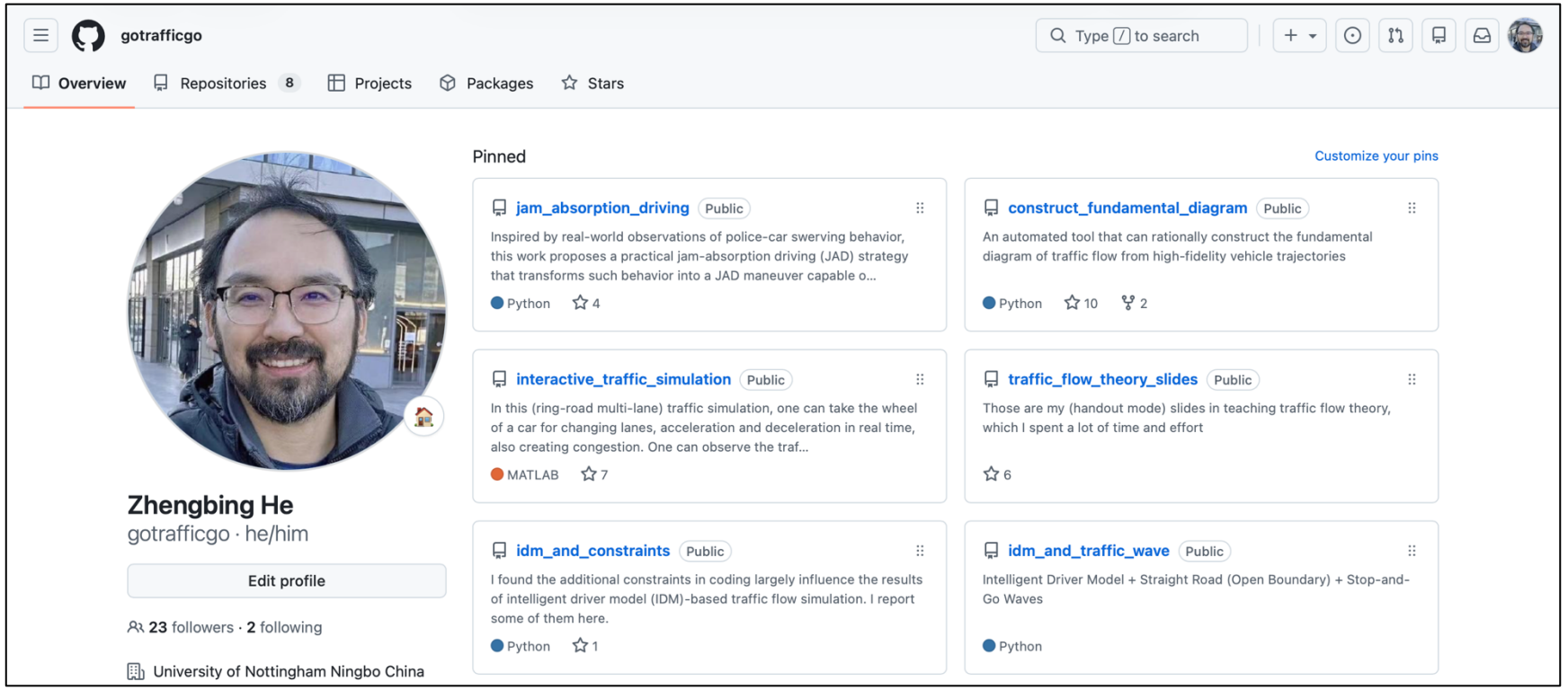




buffer

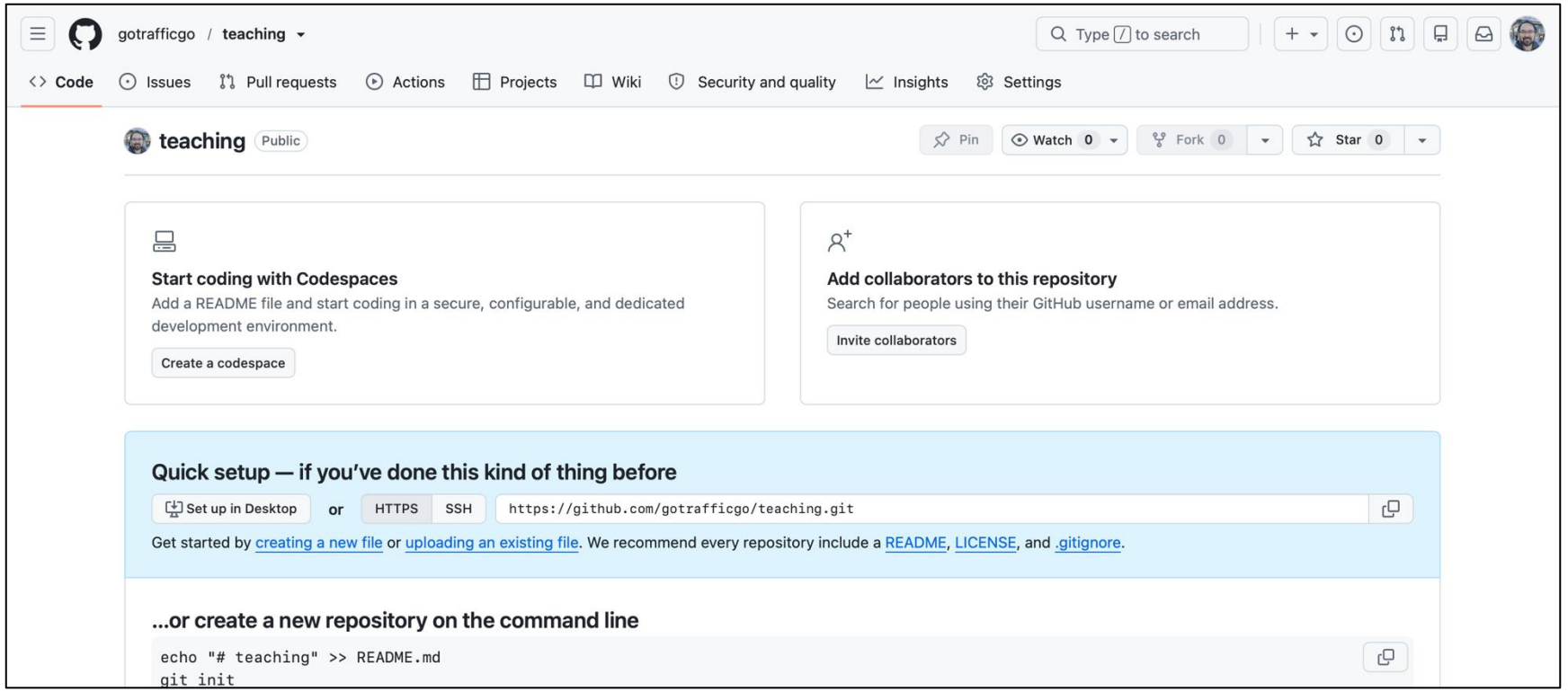
For a New Computer w/ macOS

- **GitHub and collaborate with others**
  - GitHub is a cloud-based platform for hosting Git repositories. It enables developers to store code online, collaborate with others, and manage software projects.



For a New Computer w/ macOS

- **GitHub and collaborate with others**
  - Manually create a new, **empty** repository on GitHub: *https://github.com/gotrafficgo/teaching.git*



For a New Computer w/ macOS



- **GitHub and collaborate with others**
  - **Connect local Git** with **GitHub**: *git remote add origin https://github.com/gotrafficgo/teaching.git*
  - **Verification:** *git remote –v*

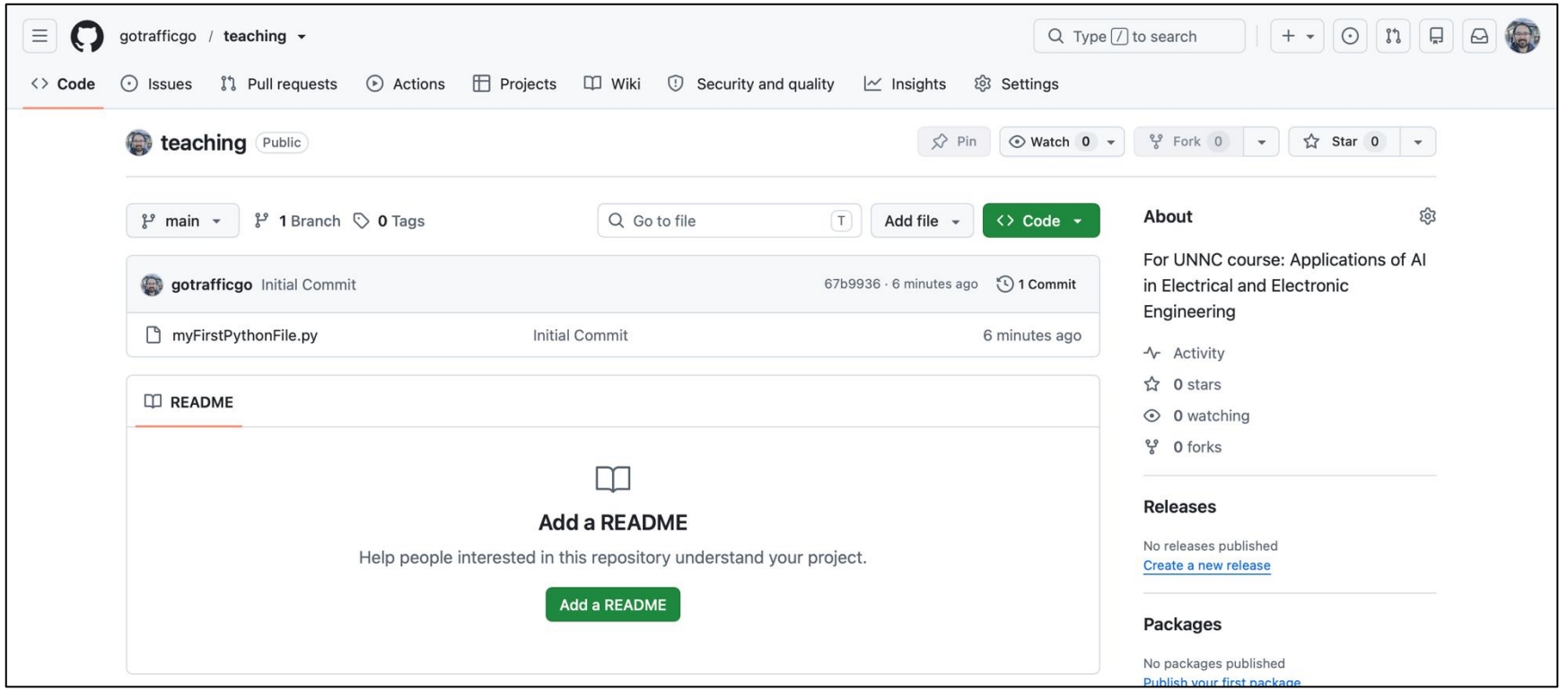
```
origin  https://github.com/gotrafficgo/teaching.git (fetch)
origin  https://github.com/gotrafficgo/teaching.git (push)
```

  - **Push local files** to **GitHub repository**: *git push -u origin main*

```
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 252 bytes | 252.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/gotrafficgo/teaching.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

For a New Computer w/ macOS

- **GitHub and collaborate with others**
  - **WELL DONE**: Others can fetch codes and push codes, too



For a New Computer w/ macOS

- **Summary: Git and GitHub**

```
git --version
git config --global init.defaultBranch main
git config --global user.name "Zhengbing He"
git config --global user.email "he.zb@hotmail.com"
git init
git status
```

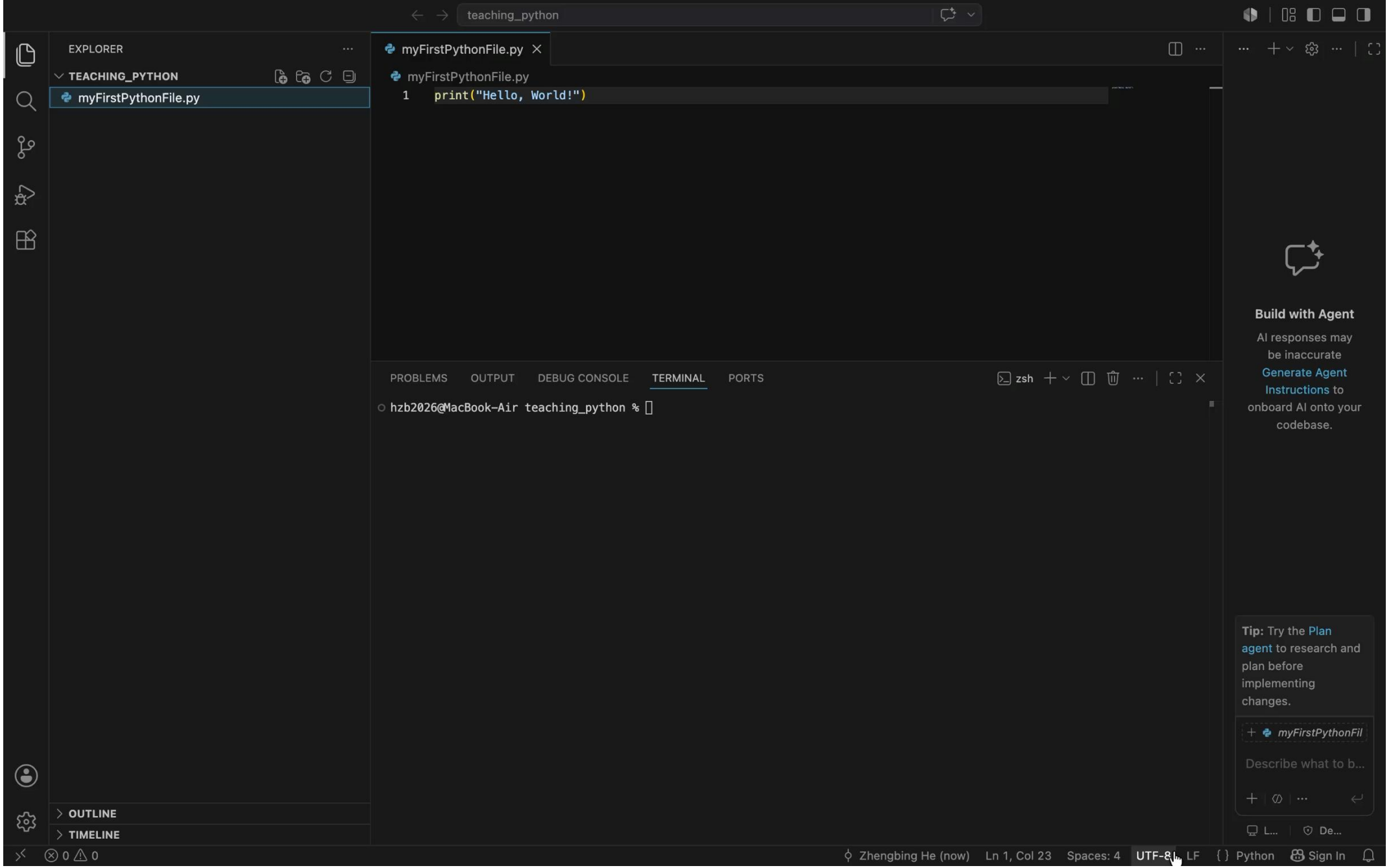
```
git remote add origin
        https://github.com/gotrafficgo/teaching.git
git remote -v

git push -u origin main
```

*git add *.py*

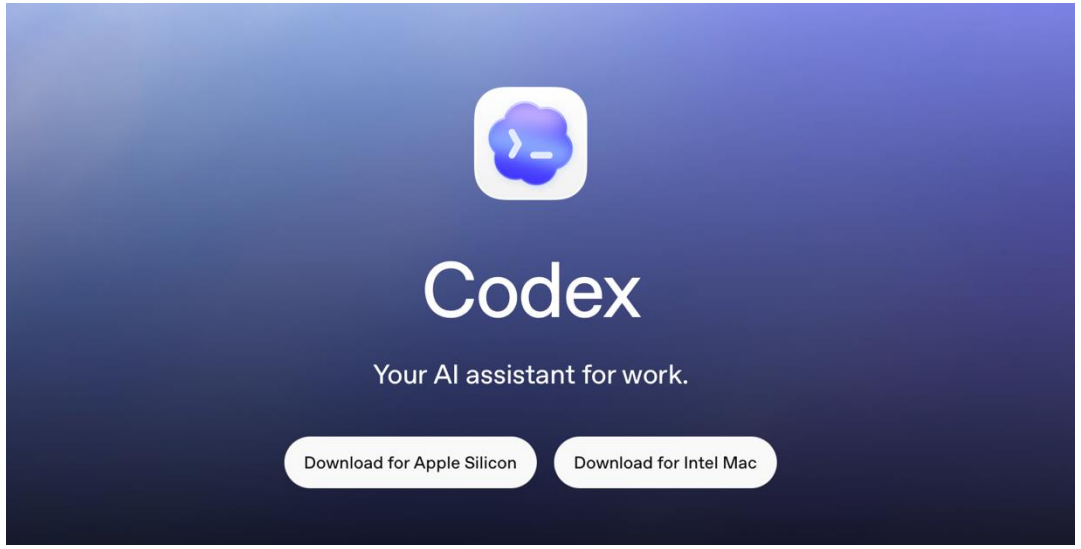
git commit -m "Initial commit"



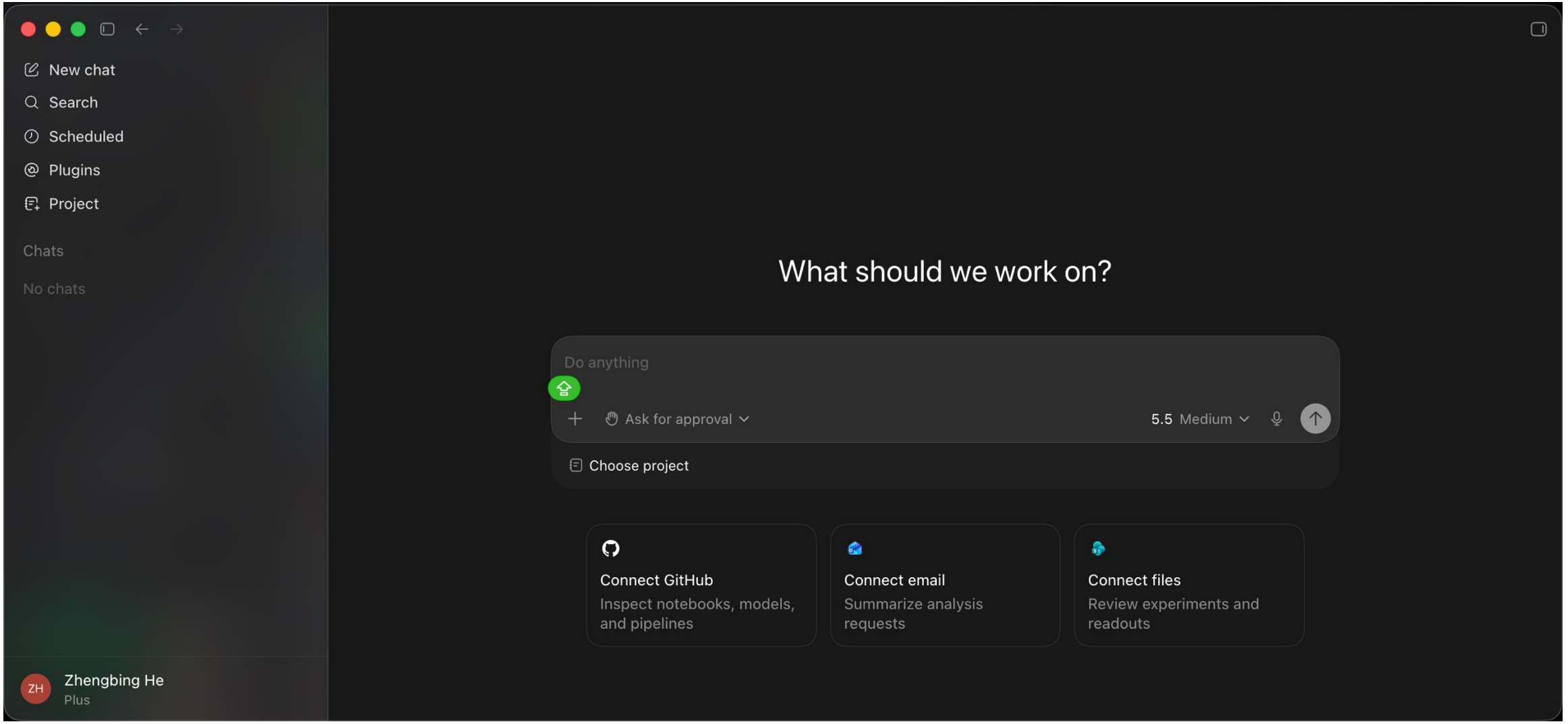

teaching_python
EXPLORER
TEACHING_PYTHON
myFirstPythonFile.py
myFirstPythonFile.py
1 print("Hello, World!")
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
zsh
hzb2026@MacBook-Air teaching_python %
OUTLINE
TIMELINE
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Tip: Try the Plan agent to research and plan before implementing changes.
myFirstPythonFil
Describe what to b...
Zhengbing He (now)
Ln 1, Col 23
Spaces: 4
UTF-8
LF
Python
Sign In

AI Agent (Codex)

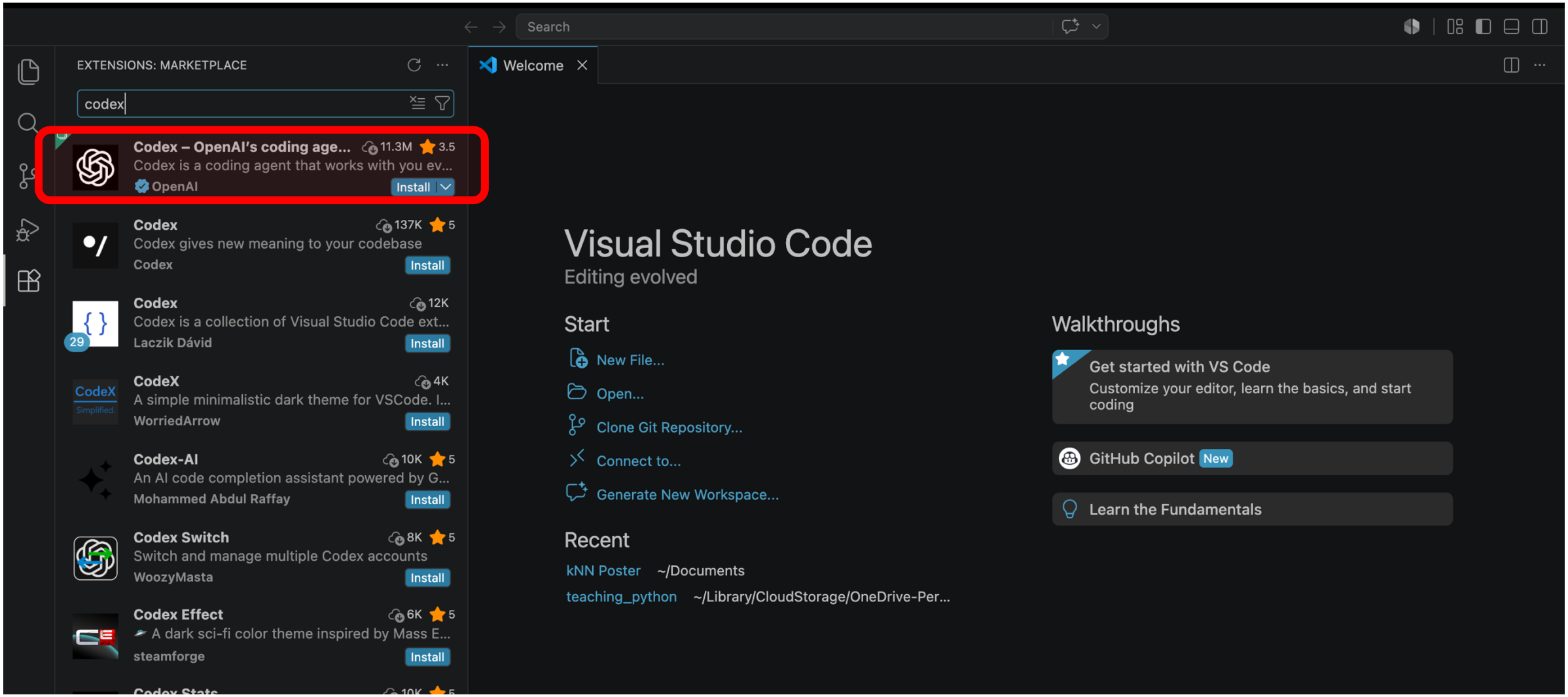
- **Codex**
  - Codex is OpenAI's AI **coding agent** that helps developers write, review, fix, and ship code
- **ChatGPT**
  - ChatGPT is OpenAI's **general-purpose AI assistant** for conversation, learning, problem solving, content creation, and everyday tasks
- **Difference**
  - ChatGPT is a broad conversational assistant, while Codex is a specialized agent designed to work directly on software-development tasks such as coding, debugging, and pull requests



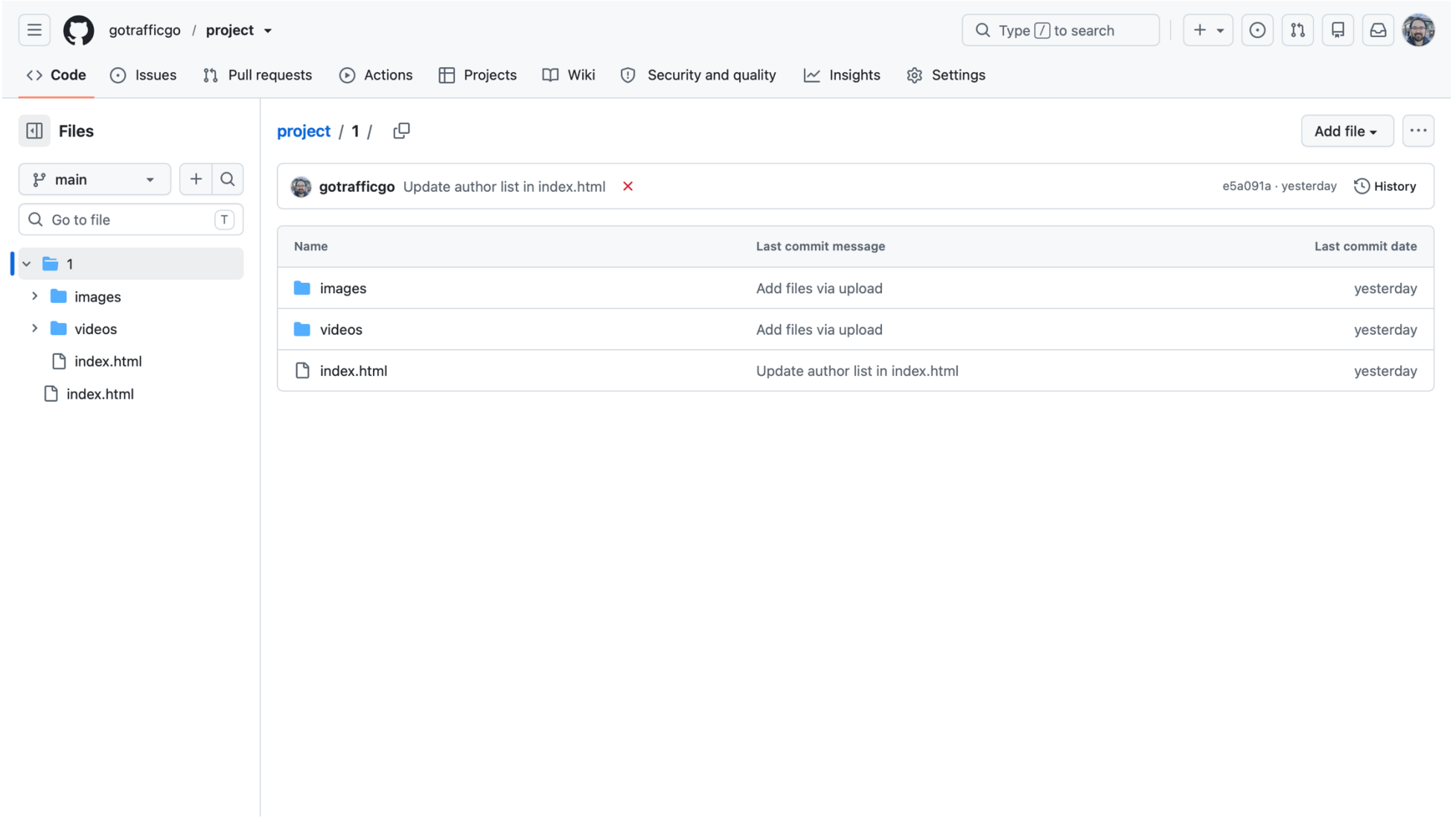
AI Agent (Codex)



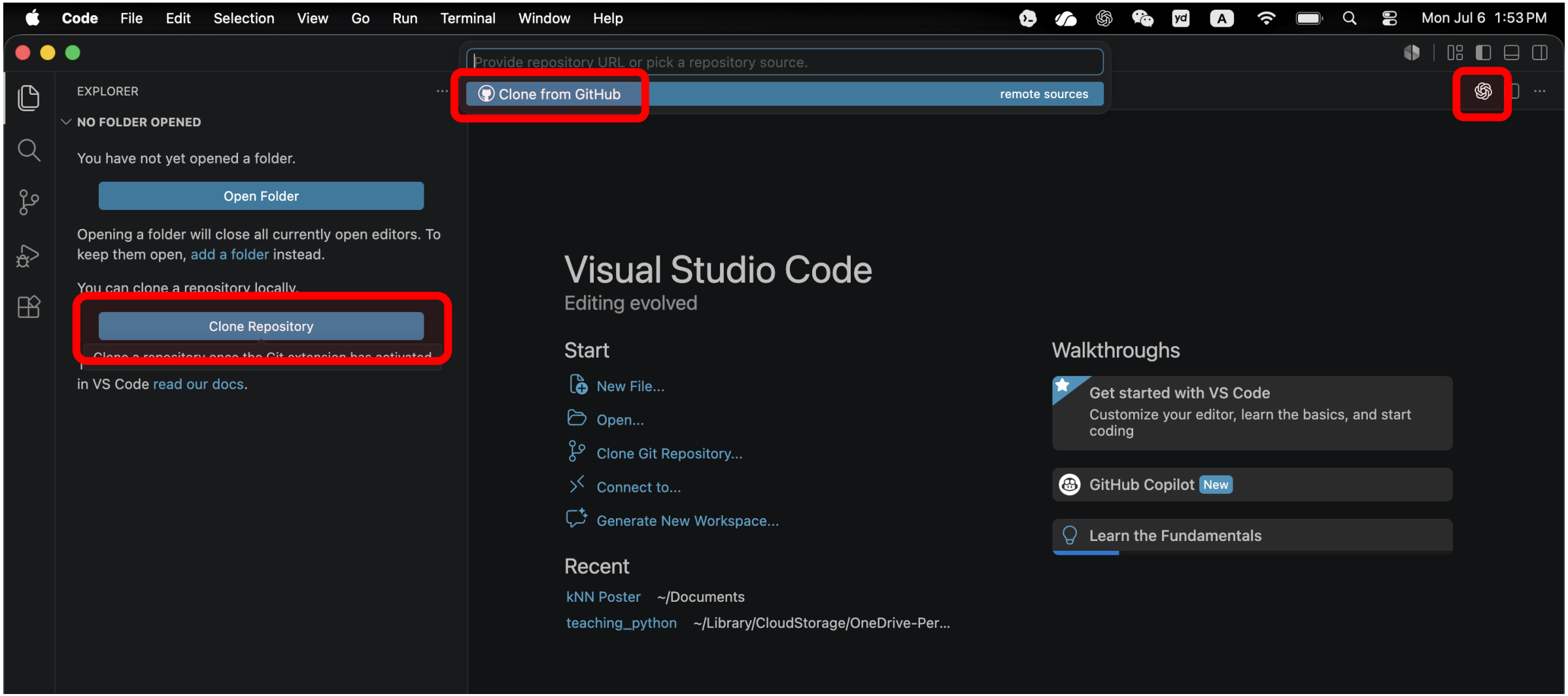
Example: VS Code + Codex



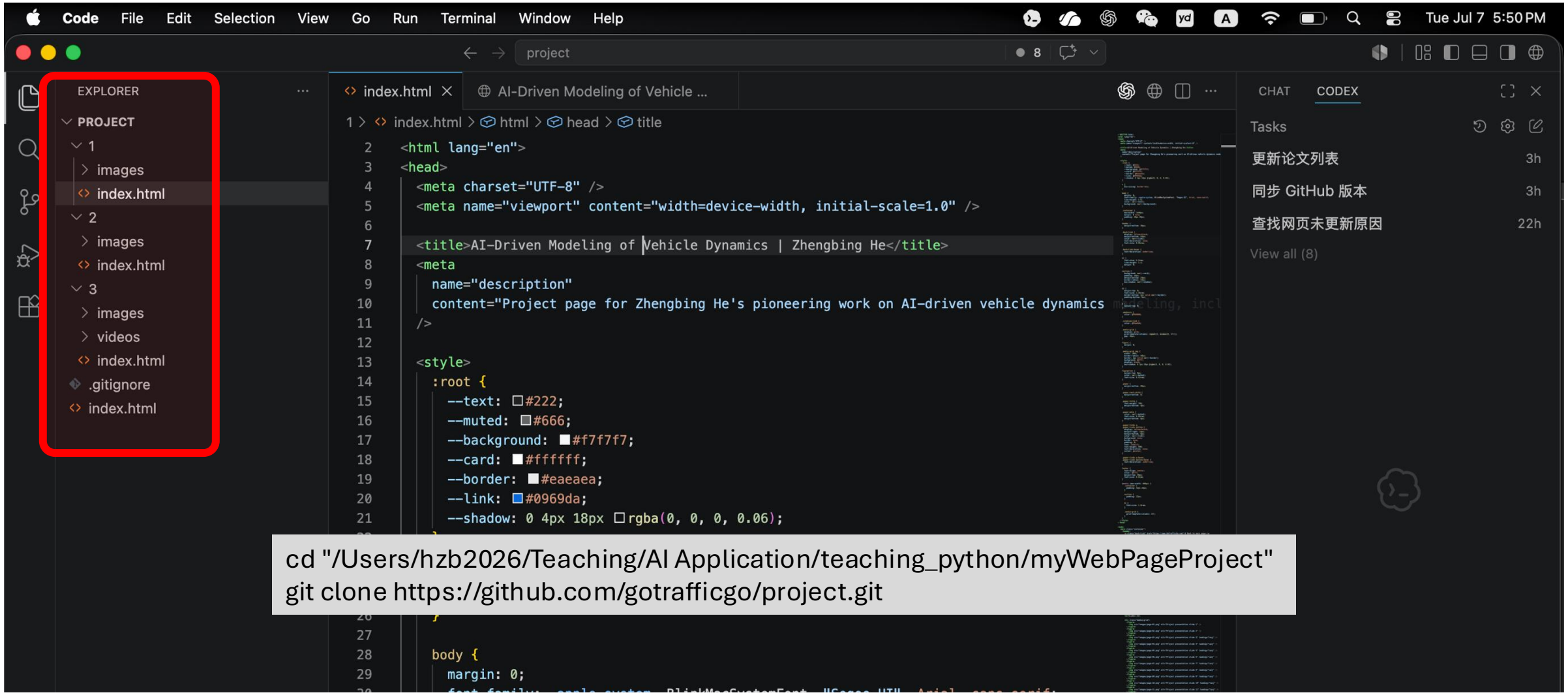
Example: Codex App



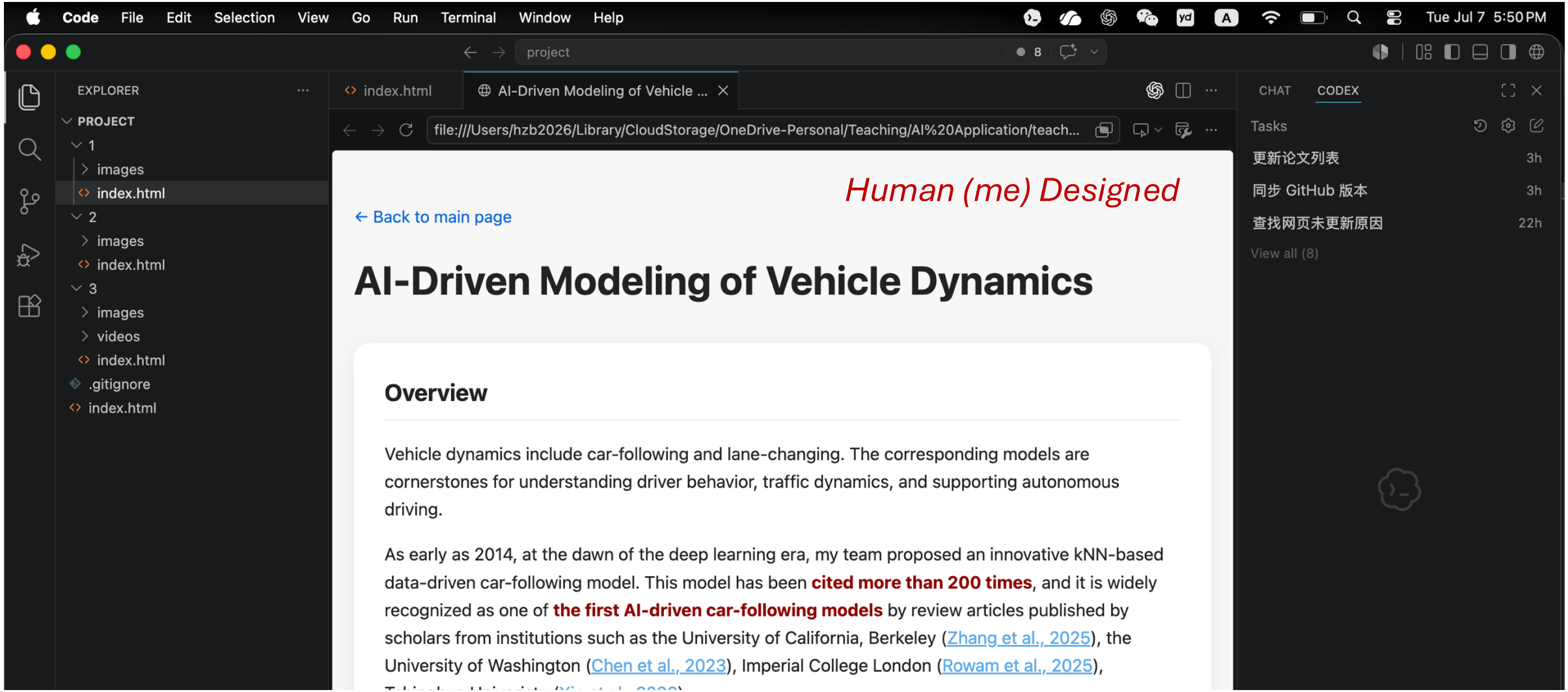
Example: VS Code + Codex



Example: VS Code + Codex



Example: VS Code + Codex



ZH Zhengbing He

← All projects

AI Designed

PROJECT 01 · INTELLIGENT TRANSPORTATION

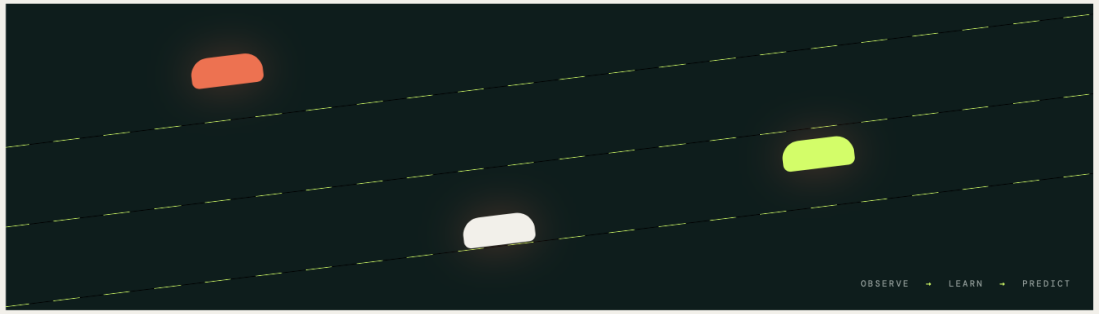
AI-Driven *Traffic Flow* Modeling

Teaching traffic models to learn from how people actually drive.

FOCUS
Driver behavior

METHODS
kNN · DBN · LSTM

TIMELINE
2014—present



01

THE WORK

From prescribed rules to learned behavior.

Car-following and lane-changing models are cornerstones for understanding driver behavior, traffic dynamics, and

THE ISSUE IS QUITE TYPICAL: FANCY, BUT NOT WHAT I WANT

Human and Professional Input Remains Critical

Coursework: Part I (20% of Final)

- **Create your own webpage on GitHub**
  - Show your experience and expertise
  - Two versions of web pages
    - Version 1: AI-generated Page, using AI Agent (Codex)
    - Version 2: You-designed Page
  - Submission: A report, showing
    - Two webpage links on the cover page, e.g.,
      - https://ThisIsTom.github.io/ai-generated-page.html
      - https://ThisIsTom.github.io/me-designed-page.html
    - How you did it
    - Your reflection (no AI usage is allowed)

Thank You

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